

Matrix Computations

Lecture 03: SVD

Hongyuan Zha

School of Data Science
The Chinese University of Hong Kong, Shenzhen

SLAI

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Outline

- 1 Geometric meaning of SVD
- 2 Algebraic derivation of SVD

Part 1. Geometric viewpoint of SVD

Singular Value Decomposition

- SVD is a useful and versatile matrix decomposition for a general rectangular matrix
- It is a bit of complicated because it involves three factors in the decomposition

$$A = U\Sigma V^T$$

where U and V are orthogonal matrices, and Σ is diagonal with nonnegative diagonal entries for a real matrix A

- A similar version exists for A being complex
- SVD is closely related to spectral decomposition of symmetric matrices
- We start with a geometric picture of SVD

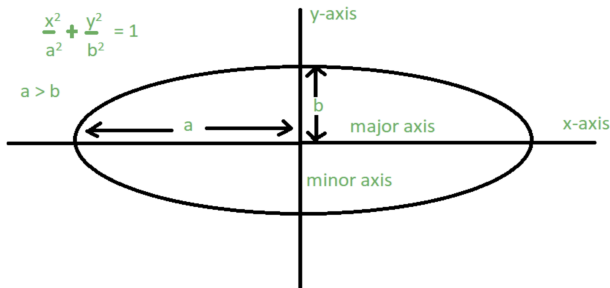
An ellipse in R^2

- In R^2 , the unit circle (or sphere in higher dimension) is denoted by

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in R^2 : x^2 + y^2 = 1 \right\}$$

- In R^2 , an ellipse is given by

$$E = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in R^2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right\}$$



Hyperellipse

- It turns out $E = \text{diag}(a, b)S$, i.e., E is the image of the unit circle under the transformation by the matrix $\text{diag}(a, b)$ with $a > 0, b > 0$
- In R^m , we can similarly define the **unit sphere**

$$S = \{u \in R^m : \|u\|_2 = 1\}$$

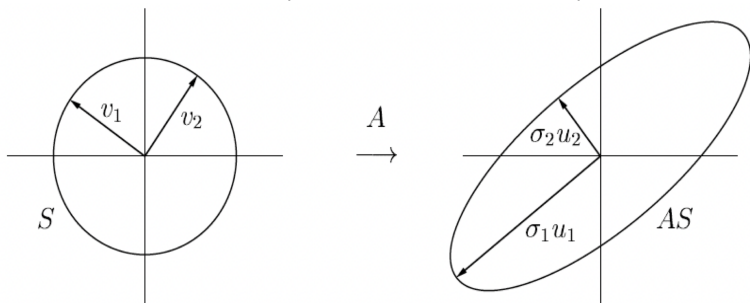
and the image under $A = \text{diag}(\sigma_1, \dots, \sigma_m)$ is an **hyperellipse**

$$E = \left\{ u \in R^m : \frac{u_1^2}{\sigma_1^2} + \dots + \frac{u_m^2}{\sigma_m^2} = 1 \right\}$$

the **principal axes** are the canonical axes, and the lengths of the principal semi-axes $\sigma_1 e_1, \dots, \sigma_m e_m$ are $\sigma_1, \dots, \sigma_m$. Here $e_1, \dots, e_m$ are the canonical axes (i.e., canonical basis vectors of R^m)

General A

- The above description is for a diagonal A
- It turns out for a general $A \in R^{m \times n}$, and S is the unit sphere in R^n , AS , the image of S under the linear mapping A , is also an hyperellipse, just that its principal axes will be **oriented** differently from the canonical axes (canonical basis vectors)



SVD from the perspective of linear transformation

- As discussed above, we consider A as a **linear mapping** from R^n (the domain of A) to R^m (the range of A). The SVD of A

$$A = U\Sigma V^T$$

can be rewritten as

$$AV = U\Sigma$$

- This implies that when we use V as an orthonormal basis of R^n and U as an orthonormal basis of R^m , A becomes the diagonal matrix Σ
- This is an example of **change of bases** for a linear transformation

Definitions for SVD

- $Av_i = \sigma_i u_i, i = 1, \dots, n$
- Given an arbitrary vector x . Let $x = \sum_{i=1}^n \alpha_i v_i$, the orthogonal expansion of x in terms of the orthonormal basis V .
- Then the image y of x under transformation A is given by

$$y = Ax = \sum_{i=1}^n \sigma_i \alpha_i u_i$$

which is the orthogonal expansion of y in terms of the orthonormal basis U

- **Conclusion.** Any matrix can be considered as diagonal, provided we pick appropriate orthogonal coordinate systems for its domain and range

Definitions for SVD

- For now let us assume $A \in R^{m \times n}$ has full-column rank (hence $m \geq n$).
- Singular values of A : the lengths of the n principal semi-axes of AS , $\sigma_1, \dots, \sigma_m$
- Left singular vectors of A : those unit vectors $u_1, \dots, u_n$ aligned with the principal semi-axes. $\sigma_i u_i$ are the principal semi-axes
- Right singular vectors of A : the *pre-image* of $\sigma_i u_i$, i.e., the unit vector v_i such that

$$Av_i = \sigma_i u_i, \quad i = 1, \dots, n$$

(Do they exist, and if yes, are they unique?)

Reduced SVD

- In a matrix equation,

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$$

which we denote as

$$AV = \hat{U}\hat{\Sigma}$$

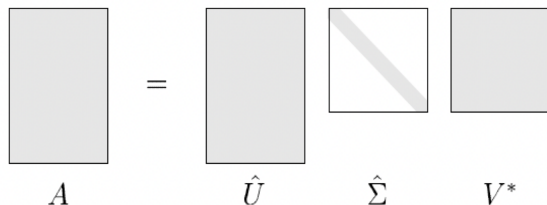
- It turns out that $V \in R^{n \times n}$ (columns of which defined as the pre-image of $\sigma_i u_i$) is orthogonal (this needs proof!), we have

$$A = \hat{U}\hat{\Sigma}V^T$$

the *reduced* SVD of A

- $A = \hat{U}\hat{\Sigma}V^T$ represented schematically as

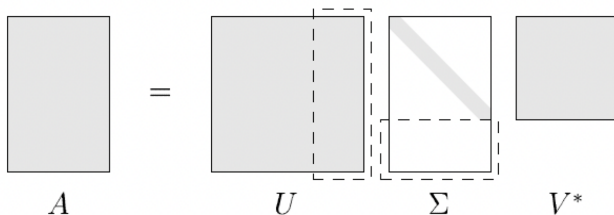
Reduced SVD ($m \geq n$)



- Since the columns of $\hat{U}$ are orthonormal, we can expand them into an orthogonal matrix $U = [\hat{U}, *] \in R^{m \times m}$ by adding orthonormal columns indicated as "*"

- We add $m - n$ rows of zeros to $\hat{\Sigma}$ to obtain $\Sigma \in R^{m \times n}$, we have the full SVD of A : $A = U\Sigma V^T$

Full SVD ($m \geq n$)



- If A is of rank $r < n$, similar argument can be used, and we end up with r nonzero singular values, and $n - r$ zero singular values

SVD: Matlab examples

X =

1 2

3 4

5 6

7 8

Matlab command

[U,S,V] = svd(X)

U =

-0.1525 -0.8226 -0.3945 -0.3800

-0.3499 -0.4214 0.2428 0.8007

-0.5474 -0.0201 0.6979 -0.4614

-0.7448 0.3812 -0.5462 0.0407

SVD: Examples

$S =$
14.2691 0
0 0.6268
0 0
0 0

$V =$
-0.6414 0.7672
-0.7672 -0.6414

SVD: Examples

```
[U,S,V] = svd(X,0)
```

```
U =
```

```
-0.1525 -0.8226
```

```
-0.3499 -0.4214
```

```
-0.5474 -0.0201
```

```
-0.7448 0.3812
```

```
S =
```

```
14.2691 0
```

```
0 0.6268
```

```
V =
```

```
-0.6414 0.7672
```

```
-0.7672 -0.6414
```

```
numpy.linalg.svd
```

```
linalg.svd(a, full_matrices=True, compute_uv=True,  
hermitian=False)
```

Part 2. Algebraic derivation of SVD

SVD and eigenvalue decomposition

- The SVD of A implies

$$A^T A = V \Sigma^T \Sigma V^T, \quad A A^T = U \Sigma \Sigma^T U^T$$

Notice that $\Sigma^T \Sigma \in R^{n \times n}$, and $\Sigma \Sigma^T \in R^{m \times m}$ are diagonal matrices, and the above give the eigenvalue decomposition of $A^T A$ and $A A^T$, respectively

- The left singular vectors are the eigenvectors of $A^T A$, and the right singular vectors are the eigenvectors of $A A^T$
- Eigenvalues of $A^T A$ are

$$\sigma_1^2, \dots, \sigma_n^2$$

- Eigenvalues of $A A^T$ are

$$\sigma_1^2, \dots, \sigma_n^2, \underbrace{0, \dots, 0}_{m-n}$$

- **Theorem.** Let $A \in \mathbb{C}^{m \times n}$, then there are unitary matrices U and V , and diagonal matrix Σ such that

$$A = U\Sigma V^H$$

where the diagonal elements of Σ are nonnegative

$$\sigma_1 \geq \cdots \geq \sigma_{\min\{m,n\}}$$

- **Proof.** Take an arbitrary nonzero eigenvalues σ^2 of $A^H A$,

$$A^H A v = \sigma^2 v, \quad \|v\|_2 = 1$$

i.e., (σ^2, v) is an eigenpair of $A^H A$

- Now define

$$u = \frac{1}{\sigma} Av$$

Then $\|u\|_2 = 1$, and

$$Av = \sigma u, \quad A^H u = \sigma v$$

We call (σ, u, v) a singular value *triplet* of A .

- Let $U = [u, \tilde{U}]$ and $V = [v, \tilde{V}]$ be unitary, we have

$$U^H AV = \begin{bmatrix} u^H Av & u^H A\tilde{V} \\ \tilde{U}^H Av & \tilde{U}^H A\tilde{V} \end{bmatrix}$$

SVD: a proof

- Now

$$u^H A v = \sigma u^H u = \sigma, \quad \tilde{U}^H A v = \sigma \tilde{U}^H u = 0, \quad u^H A \tilde{V} = 0$$

we have

$$U^H A V = \begin{bmatrix} \sigma & 0 \\ 0 & \tilde{U}^H A \tilde{V} \end{bmatrix}$$

Now use induction on $\tilde{U}^H A \tilde{V}$ to complete the proof

About the uniqueness of the singular vectors:

Lemma. Let A be Hermitian, and all the eigenvalues of A are distinct, then the eigenvectors of A are unique up to complex signs (proof in **Tutorial**)

2-norm, Frobenius norm and singular values

- Since $\|\cdot\|_2$ and $\|\cdot\|_F$ are unitarily invariant, and let $A = U\Sigma V^H$ be its SVD

$$\|A\|_2 = \|\Sigma\|_2 = \sigma_1$$

$$\|A\|_F = \|\Sigma\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_{\min\{m,n\}}^2}$$

- From the above, it is easy to see $\|A\|_2 \leq \|A\|_F$, which we proved before using Cauchy-Schwarz

SVD: geometric interpretation again

- Let us go back to the geometric meaning of SVD, and let S be the unit sphere in R^n , and $A = U\Sigma V^T$ be its SVD
- $AS = U(\Sigma(V^T S))$
 - since V is orthogonal, $V^T S$ is still the unit sphere
 - $\Sigma(V^T S)$ is an hyperellipse with the canonical basis as the set of principal semi-axes
 - $U(\Sigma(V^T S))$ then rotate and reflect this hyperellipse which gives rise to an hyperellipse with principal semi-axes given by the columns of U (corresponding to the nonzero singular values)

$$\begin{array}{ccccccc} \left[\begin{array}{c} \\ \\ \\ \end{array} \right] & = & \left[\begin{array}{c} | \\ | \\ | \\ \end{array} \right] & \left[\begin{array}{c} \diagdown \\ \Sigma \\ \diagup \\ \end{array} \right] & \left[\begin{array}{c} \text{---} V^T \text{---} \\ \end{array} \right] \\ M & & U & & & & \\ m \times n & & m \times k & & k \times k & & k \times n \\ & & & & k = \text{rank } M & & \end{array}$$

SVD: geometric interpretation again

